2025/09 PHY301 Thermodynamic and Statistical Physics

Problem set 3

Quantum statistics, Fermi and Bose gas

Problem 1: Density matrix for two-particle system[25 marks]

Consider a single harmonic oscillator with the Hamiltonian

$$\mathcal{H} = -\frac{\hbar^2}{2m}\frac{\partial}{\partial^2 x^2} + \frac{m\omega^2}{2}x^2$$

with eigenvalues $\varepsilon_n = (n + \frac{1}{2})\hbar\omega$ and eigenfunction

$$\phi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{H_n(\xi)}{(2^n n!)^{1/2}}\mathrm{e}^{-\xi^2/2}$$

where $\xi = (m\omega/\hbar)^{1/2}x$. The Hermite polynomial can be obtained by

$$H_n(\xi) = (-1)^n \mathrm{e}^{\xi^2}\frac{\partial^n}{\partial\xi^n}\mathrm{e}^{-\xi^2}$$

In canonical ensemble, the density matrix is $\rho = \mathrm{e}^{-\beta\mathcal{H}}/\mathcal{Z}$ where the partition function $\mathcal{Z} = \mathrm{tr}\mathrm{e}^{-\beta\mathcal{H}}$.

(a) Show that the density matrix in real space x-representation is

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\mathrm{e}^{-\frac{1}{2}(\xi^2+\xi'^2)}\sum_{n=0}^{\infty}\frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!}H_n(\xi)H_n(\xi')$$

(b) To evaluate the about n-sum, we can use a very useful and common trick to turn the derivative into a product of polynomials by using Fourier transformation. Namely,

$$\frac{d^n}{dx^n}f(x) = \int_{-\infty}^{\infty}(ik)^n \mathrm{e}^{ikx}\tilde{f}(k)\frac{dk}{2\pi}$$

where $\tilde{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}$ is the fourier transformation of $f(x)$. With this, prove

the following

$$\sum_{n=0}^{\infty}\frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!}H_n(\xi)H_n(\xi') = \frac{\mathrm{e}^{-\frac{1}{2}\beta\hbar\omega}\mathrm{e}^{\xi^2+\xi'^2}}{4\pi}\int dk\int dk'\exp[-\frac{1}{4}(k^2+2kk'\mathrm{e}^{-\beta\hbar\omega}+k'^2)+i(k\xi+k'\xi')] \tag{1}$$

(c) The above integration is the *generalized* version of Gaussian integral in higher dimensional vector space. For instance, in a n-dimensional vector space,

$$\int dx^n \mathrm{e}^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x}+\mathbf{x}^T\mathbf{J}} = \sqrt{\frac{(2\pi)^n}{|\mathbf{A}|}}\mathrm{e}^{-\frac{1}{2}\mathbf{J}^T\mathbf{A}^{-1}\mathbf{J}} \tag{2}$$

where the $n\times n$ matrix $[\mathbf{A}]_{ij}=A_{ij}$, the vectors $\mathbf{x}=[x_1,\ldots x_n]$ and $\mathbf{J}=[J_1,\ldots,J_n]$. [*Note.* Gaussian integration in n-dimensional vector space is an important integration that we need to know in physics. You can derive or find the proof by yourself.]. Using Eq.(2), we therefore integrat out k and k' in Eq.(1), and show that the density matrix is

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \sqrt{\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}}\exp\Big[-\frac{m\omega}{4\hbar}\Big((x+x')^2\tanh\frac{\beta\hbar\omega}{2}+(x-x')^2\coth\frac{\beta\hbar\omega}{2}\Big)\Big]$$

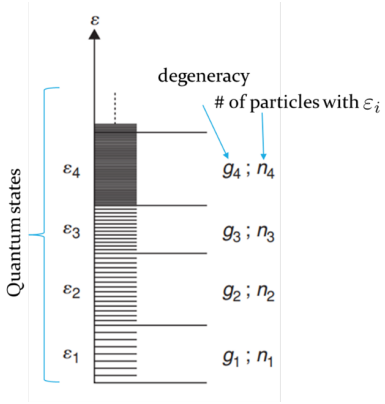
(d) Calculate the partition function and density matrix

(e) Using the density matrix, calculate $\langle x^2\rangle$

(f) Using the density matrix, calculate $E=\langle H\rangle$.

Problem 2: Quantum statistics [25 marks]

The one-particle's energy spectrum is demonstrated in the figure below.



The N-particle Hamiltonian is

$$\mathcal{H}_N(\{n_{i\alpha}\}) = \sum_{i=1} \sum_{\alpha=1}^{g_i} \varepsilon_i n_{i\alpha} \tag{3}$$

Here, $n_{i\alpha}$ is the occupation number of particles in state α with energy ε_i and $n_i = \sum_{\alpha=1}^{g_i} n_{i\alpha}$. In canonical ensemble with N identical particles, the partition function is $\mathcal{Z}_N = \sum_{\{n_{i\alpha}\}} \mathrm{e}^{-\beta \mathcal{H}_N(\{n_{i\alpha}\})}$ The sum is subject to the constraint of total number of particles $\sum_i \sum_{\alpha=1}^{g_i} n_{i\alpha} = N$. To relax this constraint, we use grand canonical ensemble to model our quantum system and the grand partition

$$\mathcal{Q} = \sum_{N=0}^{\infty} \mathrm{e}^{\beta \mu N} \mathcal{Z}_N$$

(a) Calculate the Grand partition function for *Boson* and *Fermion*.

(b) Using the result in (a), calculate the mean occupation number $\langle n_i \rangle = \langle \sum_{\alpha_i=1}^{g_i} n_{i\alpha} \rangle$ for both bosonic and fermionic system

(c) What is the probability of finding $n_{i\alpha}$ particles in state α with energy ε_i.

Problem 3: Bose Gas with arbitrary dispersion in d-dimension [25 marks]

consider a gas of non-interacting (spinless) bosons with energy $\varepsilon = cp$ ($p = |\mathbf{p}|$), contained in a box of volume $V = L^3$.

(a) Calculate the grand potential $\mathcal{G} = -k_B T \ln \mathcal{Q}$, and the density $n = N/V$, at a chemical potential μ. Express your answers in terms of Bose-Einstein function $g_\nu(z)$,

where $z = \mathrm{e}^{\beta\mu}$, and

$$g_\nu(z) = \frac{1}{(\nu-1)!}\int_0^\infty \frac{x^{\nu-1}dx}{z^{-1}\mathrm{e}^x - 1}$$

(b) Calculate the gas pressure P, its energy U, and compare the ratio U/PV to the classical value.

(c) Find the critical temperature, $T_c(n)$, for Bose–Einstein condensation.

(d) What is the temperature dependence of the heat capacity C_V for $T < T_c(n)$?

(e) Evaluate the dimensionless heat capacity C_V/Nk_B at the critical temperature $T = T_c$, and compare its value to the classical (high-temperature) limit.

Problem 4: Non-interacting fermions [25 marks]

Consider a grand canonical ensemble of non-interacting fermions in the two-dimension crystal with chemical potential μ. The one-particle states are labeled by a wave vector $\vec{k}$, and have energies $\varepsilon_{\vec{k}} = \hbar^2k^2/(2m)$.

(a) What is the Fermi energy ε_F compare to 3D case?

(b) What is the chemical potential $\mu(T)$? [in this case you can find the exact expression for $\mu(T)$.]

(c) Calculate the grand potential function and find the ratio PV/U where P, V, and U are the pressure, volume and internal energy.

(d) What is the heat capacity C_V compare to the 3D case.

(e) Calculate the probability for a particle to *occupy* the state with energy $\varepsilon = \mu + \delta$ (above chemical potential) and the probability to *unoccupy* the states with energy $\varepsilon = \mu - \delta$ (below chemical potential). Do these two probability the same?